AML TRANSACTION & FRAUD ANALYSIS

Page 1 – Introduction

1. Introduction

Anti-Money Laundering (AML) is the process of identifying, preventing, and reporting financial activities that may involve illegally obtained money. Financial institutions such as banks, insurance companies, and other financial service providers use AML systems to monitor transactions and detect suspicious activities.

The **AML Transaction & Fraud Analysis Dashboard** is a data analysis and visualization project designed to examine financial transactions, identify potential fraud, monitor transaction amounts, and analyze suspicious alert patterns. The dashboard presents important information through key performance indicators (KPIs), charts, and interactive filters.

Money laundering and financial fraud are major challenges in the financial industry. Criminals may attempt to hide the origin of illegal money by transferring it through multiple accounts or conducting unusual transactions. Manual monitoring of large numbers of transactions can be time-consuming and difficult. Therefore, data analytics and automated monitoring tools are useful for identifying unusual patterns.

2. Objectives of the Project

The main objectives of the AML Transaction & Fraud Analysis Dashboard are:

1. To analyze the total value of financial transactions.
2. To monitor the number of transactions and accounts.
3. To identify transactions classified as potentially fraudulent.
4. To compare fraudulent and non-fraudulent transaction amounts.
5. To analyze alert categories such as Cycle and Fan-in.
6. To observe transaction activity over time.
7. To support financial institutions in investigating suspicious transactions.

3. Importance of AML Analysis

AML analysis helps financial institutions understand customer transaction behavior and identify activities that require additional investigation. It can support regulatory compliance, reduce financial risks, and improve the efficiency of fraud-monitoring processes.

The dashboard provides a visual summary of transaction data, making it easier for analysts and decision-makers to understand financial activity without examining every record individually.

Page 2 – Dashboard Overview and KPI Analysis

4. Overview of the Dashboard

The dashboard titled “AML Transaction & Fraud Analysis” contains KPI cards, a doughnut chart, horizontal and vertical bar charts, a line chart, and filters.

The main dashboard indicators include total transaction amount, average transaction amount, total initial balance, total accounts, total transactions, and fraudulent transactions.

5. Key Performance Indicator Analysis

5.1 Total Transaction Amount

The dashboard displays a total transaction amount of approximately **153.48 billion**. This indicator represents the overall monetary value of the transactions included in the dataset. It helps analysts understand the scale of financial activity being monitored.

5.2 Total Transactions

The dashboard displays approximately **1 million transactions**. A large transaction dataset requires systematic analysis because unusual transactions may be difficult to identify through manual checking.

5.3 Total Accounts

The dashboard shows approximately **10,000 accounts**. Analyzing transaction activity across multiple accounts may help identify unusual relationships, repeated transfers, and patterns that require investigation.

5.4 Average Transaction Amount

The average transaction amount is displayed as approximately **115.99K**. This measure represents the average monetary value per transaction, according to the dashboard's calculation.

5.5 Total Initial Balance

The total initial balance is shown as approximately **2.80 million**. This indicator provides information about the initial balances recorded in the dataset. The exact meaning depends on how the underlying data model defines and aggregates the balance field.

5.6 Fraudulent Transactions

The dashboard displays approximately **2,000 fraudulent transactions**. These are transactions classified as fraudulent in the dataset. They should be treated as flagged or labelled cases rather than independently confirmed fraud unless verification has been completed.

These KPIs provide an overview of transaction activity and help analysts decide which areas need closer examination.

Page 3 – Analysis of Charts and Transaction Patterns

6. Total Transaction Amount by Fraud Status

The doughnut chart compares the total transaction amount for two fraud categories: True and False. The chart displays an overall transaction amount of approximately 153.48 billion.

This visualization helps analysts compare the monetary value associated with transactions in each category. A transaction's monetary value is different from its frequency, so both amount-based and count-based comparisons are important.

For example, a category containing fewer transactions may still have a high total value if its individual transactions are large.

7. Count of Alert ID by Alert Type

The horizontal bar chart compares the number of alerts associated with two alert types: Cycle and Fan-in.

The dashboard indicates that the Cycle category has a higher alert count than the Fan-in category.

A Cycle pattern may refer to transactions that form a circular transfer path between accounts. A Fan-in pattern may refer to funds arriving from multiple accounts into one account. Such patterns can occur in legitimate financial activity as well as suspicious activity.

Therefore, these alert categories can help analysts identify patterns for further review, but they do not independently prove that money laundering has occurred.

8. Total Transactions by Timestamp

The line chart shows transaction activity across timestamps. The displayed pattern rises sharply at the beginning and then fluctuates within a relatively stable range.

Time-based analysis helps analysts understand when transaction activity increases or decreases. Sudden spikes, unusual transaction frequency, or activity outside a customer's normal pattern may require additional investigation.

Comparing transaction activity across different periods can also help identify repeated patterns and unusual changes in financial behavior.

9. Transaction Amount Comparison by Fraud Status

The final bar chart compares transaction amounts for fraudulent and non-fraudulent categories. The non-fraudulent category has a substantially higher bar than the fraudulent category.

This indicates that the monetary value of non-fraudulent transactions is higher in the displayed comparison. However, the chart alone does not establish the exact amount of fraud or the percentage of transactions that are fraudulent.

Page 4 – Filters, Findings, and Interpretation

10. Interactive Dashboard Filters

The dashboard includes filters that allow users to narrow the displayed results.

10.1 IS_FRAUD Filter

This filter allows users to select transactions classified as True or False. It can be used to examine the characteristics of fraudulent and non-fraudulent transaction groups separately.

10.2 ALERT_TYPE Filter

This filter provides alert categories such as Cycle and Fan-in. Selecting a category allows analysts to focus on its associated transaction activity and compare alert patterns.

These filters improve the usefulness of the dashboard because users can explore specific groups instead of viewing all records at once.

11. Major Findings

Based on the displayed dashboard, the following findings can be identified:

1. The dataset contains approximately one million transaction records.
2. The total transaction amount is approximately 153.48 billion.
3. The dashboard represents approximately 10,000 accounts.
4. The average transaction amount is displayed as 115.99K.
5. Approximately 2,000 transactions are labelled as fraudulent.
6. Cycle alerts are more numerous than Fan-in alerts in the displayed chart.
7. Transaction activity changes over time, making timestamp analysis useful.
8. The total amount associated with non-fraudulent transactions is higher than that associated with fraudulent transactions in the final bar chart.

These findings summarize the visual information available in the dashboard. The underlying dataset is required to verify exact calculations and determine the reasons behind the observed patterns.

12. Limitations of the Analysis

The dashboard provides an overview rather than a complete investigation of every suspicious transaction. The screenshot does not reveal the detailed transaction records, customer profiles, exact alert definitions, or model performance.

A flagged transaction may be legitimate, while a fraudulent transaction may not always be detected. Consequently, additional checks, appropriate thresholds, and investigation procedures are necessary before taking action.

Page 5 – Recommendations and Conclusion

13. Recommendations

The following improvements can increase the effectiveness of AML transaction monitoring.

13.1 Monitor High-Risk Transactions

Transactions involving unusual amounts, repeated transfers, or unexpected account activity should be reviewed according to established risk criteria.

13.2 Improve Alert Investigation

Cycle and Fan-in alerts should be examined alongside transaction history, account relationships, and relevant customer information. This can help analysts distinguish normal activity from suspicious patterns.

13.3 Use Time-Based Monitoring

Transaction activity should be monitored continuously or at suitable intervals. Sudden increases in transaction frequency or value can be highlighted for further investigation.

13.4 Reduce False Positives

Monitoring rules should be reviewed regularly to reduce unnecessary alerts while maintaining appropriate detection of suspicious activity. Any automated classification should be evaluated against reliable labelled data.

13.5 Improve Data Quality

Transaction records, timestamps, account identifiers, and fraud labels should be checked for missing values, duplicates, and inconsistent formats. Accurate data is essential for reliable analysis.

13.6 Introduce Advanced Analytics

Where suitable data is available, statistical analysis and machine-learning models can help identify unusual patterns. These methods should be tested, monitored, and used alongside human investigation rather than treated as infallible decision-makers.

14. Conclusion

The AML Transaction & Fraud Analysis Dashboard is a useful data visualization tool for understanding financial transactions and identifying areas that may require further investigation. It combines KPI cards, transaction charts, alert comparisons, timestamp analysis, and interactive filters to present financial information in an understandable format.

The dashboard highlights important measures such as total transaction amount, transaction count, account count, average transaction amount, and fraudulent transaction labels. It also shows that Cycle alerts are more frequent than Fan-in alerts in the displayed data.

Overall, the project demonstrates how data analytics can support anti-money laundering activities and fraud-risk management. By combining accurate data, effective monitoring rules, time-based analysis,

and appropriate investigation procedures, financial institutions can improve their ability to identify suspicious activity and make informed decisions.